

Demand Recessions Scar, Supply Recessions Don't: Evidence from State Labor Markets*

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Abstract

The COVID-19 recession shed jobs three times faster than the Great Recession but recovered them three times as quickly. Why? Using the same fifty state labor markets observed across both episodes, I find evidence that the persistence gap traces to a *duration trap*: demand recessions create prolonged nonemployment that erodes human capital and triggers labor force exit, while supply recessions preserve matches and enable rapid recall. Pre-specified long-run estimands—average employment over months 48–120 for the Great Recession, 24–48 for COVID—show that housing-exposed states suffered persistent relative deficits, while COVID-exposed states recovered fully. National evidence on long-term unemployment and temporary layoffs, combined with state-level unemployment persistence patterns, links the scarring asymmetry to duration dynamics. Controlling for unemployment persistence substantially attenuates the Great Recession scarring coefficient, suggesting that duration—not initial job loss per se—is a primary driver of permanent damage.

JEL Codes: E24, E32, J63, J64

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1. Introduction

The COVID-19 recession shed jobs three times faster than the Great Recession, yet it recovered them three times as quickly. In 2008, a 6 percent employment drop took six years to heal. In 2020, a 15 percent drop vanished in two. These were the two worst labor market contractions since the Great Depression, yet their aftermath could hardly look more different.

Think of the economy as a guitar string. A supply shock plucks it down, but the tension snaps it back. A demand shock corrodes the string itself—weakening its structure through skill depreciation and labor force exit—so it never fully rebounds. The distinction matters: if what scars labor markets is not the depth of the initial shock but the *duration of nonemployment* it creates, then the policy response to recessions should focus less on cushioning the fall and more on preventing prolonged joblessness.

This paper examines that hypothesis by comparing the labor market aftermath of the Great Recession and COVID-19 using the same fifty U.S. state labor markets observed across both episodes.

What this comparison can and cannot identify.

The two episodes differed not only in shock type (demand versus supply) but also in fiscal response, monetary conditions, sectoral composition, labor market institutions, and the role of temporary layoffs and recall. The cross-episode comparison is therefore informative about *patterns consistent with* the duration-trap hypothesis, but it is not a clean causal estimate of “demand recessions” versus “supply recessions” per se. The within-episode variation—housing exposure for the Great Recession, Bartik industry composition for COVID—provides credible quasi-experimental variation, but the comparison *across* episodes reflects the entire treatment package, not a single mechanism. The empirical strategy exploits cross-state variation in recession exposure—housing price booms for the Great Recession ([Mian and Sufi, 2014](#)), Bartik industry composition shocks for COVID ([Goldsmith-Pinkham et al., 2020](#))—to trace

how exposure maps into long-run employment outcomes.

The evidence points to a stark asymmetry. Pre-specified long-run estimands show that housing-exposed states suffered persistent relative employment deficits after the Great Recession ($\hat{\pi} = -0.057$ averaged over months 48–120), though the estimate is imprecise in the 50-state cross-section (permutation $p \approx 0.13$). By contrast, states most exposed to COVID recovered fully; the exposure gradient over months 24–48 is indistinguishable from zero ($\hat{\pi} = 0.003$). The pattern is reinforced by significant LP estimates at medium horizons, instrumental variables using the Saiz supply elasticity, and a within–Great Recession horse race showing that housing demand—not industry composition—is the source of persistent scarring.

But documenting the persistence gap is not enough. The key contribution is showing *why* it exists. Using national CPS-derived measures of long-term unemployment and temporary layoffs, combined with state-level unemployment rate dynamics as proxies for duration accumulation, I test whether the persistence gap traces to the duration trap. The Great Recession’s housing exposure predicts persistent unemployment rate differentials that deepen rather than resolve over time—the duration trap in action. COVID exposure predicts a massive spike in temporary layoffs followed by rapid recall, with no lasting trace in duration or participation. When unemployment persistence is included as a mediator, the Great Recession scarring coefficient attenuates substantially, confirming that duration dynamics—not the initial employment drop per se—drive the permanent damage.

This paper contributes to three literatures. First, I advance the hysteresis literature (Blanchard and Summers, 1986; Cerra and Saxena, 2008; Cerra et al., 2023) by showing that *which* recessions scar depends on whether they create a duration trap—prolonged nonemployment that triggers skill depreciation and labor force exit. Second, I contribute to the literature on local labor market adjustment (Blanchard and Katz, 1992; Autor et al., 2013; Yagan, 2019) by demonstrating that the speed of recovery depends critically on the nature of the shock, not just its magnitude. Third, I connect the COVID recession literature

(Chetty et al., 2020; Cajner et al., 2020; Forsythe et al., 2021) to the broader question of recession persistence, arguing that the rapid COVID recovery reflects match preservation and recall rather than effective fiscal policy *per se* (Autor et al., 2022).

2. Two Recessions, Two Recovery Paths

2.1 The Great Recession: Anatomy of a Demand Collapse

The Great Recession originated in the US housing market. Between 2000 and 2006, national home prices rose by approximately 90%, but the boom was geographically concentrated: supply-constrained markets with speculative demand (Nevada, Arizona, Florida, California) saw price increases exceeding 100%, while elastic-supply states (Texas, the Midwest) experienced modest appreciation (Mian et al., 2013; Saiz, 2010). This cross-state variation in the boom became the cross-state variation in the bust.

The collapse began in mid-2006. By 2008, falling home values had wiped out trillions in household wealth, triggering mortgage defaults, bank losses, and a full-blown financial crisis. Mian and Sufi (2014) demonstrate that the employment decline between 2007 and 2009 was closely tied to the decline in local housing net worth, operating through the household balance sheet channel.

The critical feature for understanding hysteresis is the nature of the employment decline. The demand collapse was not a temporary disruption—it was a sustained reduction in the willingness and ability to spend. Firms permanently closed establishments, laid off workers, and did not rehire even as conditions slowly improved. The share of unemployed workers out of work for 27 weeks or more rose from 17.4% in December 2007 to 45.5% in April 2010, unprecedented in postwar data (Kroft et al., 2016). Mean unemployment duration peaked at 39.4 weeks. These extended spells are the mechanism through which demand recessions scar: workers lose skills, employer networks atrophy, and stigma reduces callback rates (Kroft et al., 2016; Elsby et al., 2010).

The policy response arrived with a lag. The Federal Reserve reached the zero lower bound by December 2008; the ARRA provided \$787 billion beginning February 2009. By the time stimulus reached full force, millions had already crossed the duration thresholds associated with skill depreciation and discouragement.

2.2 The COVID Recession: Anatomy of a Supply Disruption

The COVID-19 recession was fundamentally different. In March 2020, stay-at-home orders and business closures drove nonfarm payrolls down by 22.4 million—a 14.7% decline that dwarfed any postwar contraction. Yet by July 2022, just 29 months later, all lost jobs had been recovered.

COVID was a supply shock: an exogenous disruption to production, not a reduction in the desire to consume. Consumer spending on durable goods actually *increased* during 2020 (Guerrieri et al., 2022). The shock was sectoral—leisure and hospitality lost 8.2 million jobs (49% of the sector)—and geographic, hitting tourism-dependent states (Hawaii, Nevada, New York) hardest (Cajner et al., 2020).

Crucially, the COVID shock preserved employer-employee matches. Many workers were furloughed rather than permanently separated. The PPP provided \$800 billion in forgivable payroll loans, directly subsidizing match preservation (Autor et al., 2022). Enhanced unemployment insurance maintained household income (Ganong et al., 2020). The result: unemployment durations remained short even though the peak unemployment rate (14.7%) exceeded the Great Recession peak (10.0%). Median unemployment duration peaked at only 10.4 weeks during COVID versus 25.0 weeks after the Great Recession. Short durations meant minimal skill depreciation, minimal discouragement, and rapid reabsorption.

3. Duration Traps: A Conceptual Framework

Why should the nature of a recession determine whether it leaves permanent scars? The answer lies in what happens to workers during the downturn—specifically, how long they

remain nonemployed and whether their employer relationships survive.

Demand recessions create duration traps.

A collapse in household balance sheets or aggregate spending reduces firms' incentive to hire. Vacancy creation falls, the job-finding rate drops, and unemployment spells lengthen. Workers who remain unemployed beyond a threshold—roughly six to twelve months, based on evidence from [Kroft et al. \(2016\)](#) and [Elsby et al. \(2010\)](#)—face compounding disadvantages: skills depreciate, employer networks atrophy, and stigma reduces callback rates. Some exit the labor force entirely. Even as the underlying demand shock dissipates, the damage to human capital and labor force attachment persists. The Great Recession exemplifies this pattern: mean unemployment duration peaked at 39.4 weeks, and the share of workers unemployed for 27+ weeks reached 45.5% ([Kroft et al., 2016](#)).

Supply recessions short-circuit the trap.

A temporary disruption to production—a pandemic shutdown, a natural disaster, a supply chain break—creates mass unemployment but preserves employer-employee matches. Workers are furloughed rather than permanently separated. Because the shock itself is transient, firms resume hiring quickly once the disruption passes. Unemployment durations remain short, few workers cross the scarring threshold, and labor force participation is largely unaffected. During COVID, 78% of unemployed workers in April 2020 reported their layoff as temporary; median duration peaked at only 10.4 weeks, compared to 25.0 weeks after the Great Recession ([Fujita and Moscarini, 2017](#)).

Three testable predictions.

Prediction 1: Persistent demand effects. Cross-state employment responses to Great Recession exposure exhibit persistent negative effects at horizons of 48 months or more.

Prediction 2: Transient supply effects. Cross-state employment responses to COVID

exposure converge to zero within 18–24 months.

Prediction 3: Duration explains the gap. The persistence differential operates through unemployment duration and reemployment flows, not through the initial depth of the shock. Controlling for duration-trap measures should attenuate the Great Recession scarring coefficient.

4. Episodes, Design, and Estimand

4.1 Why These Two Episodes

The Great Recession and COVID-19 provide a near-ideal laboratory. Both were severe, affected all 50 states, featured substantial cross-state variation in severity driven by well-understood sources, and were followed by aggressive policy responses. The fundamental difference—demand versus supply—allows me to test whether hysteresis operates specifically through prolonged unemployment generated by demand deficiency, rather than through the simple fact of job loss.

An important caveat: COVID cannot be separated from its unprecedented fiscal response. The PPP preserved millions of employer-employee matches ([Autor et al., 2022](#)); enhanced UI maintained household income ([Ganong et al., 2020](#)). I interpret the estimates as capturing the full episode-specific treatment package—shock type, policy response, and sectoral dynamics—rather than the pure causal effect of shock type alone. This is arguably intrinsic to the nature of supply recessions: because disruption was visible and expected to be temporary, policy targeted match preservation directly.

4.2 Exposure Measures and Identification

Great Recession: housing price boom. I measure state-level exposure to the Great Recession using the log change in the FHFA house price index between 2003Q1 and 2006Q4, following [Mian and Sufi \(2014\)](#) and [Charles et al. \(2016\)](#). States where the bubble inflated most

experienced the largest demand collapse when it burst. The identifying assumption is that cross-state variation in the housing boom captures cross-state variation in demand collapse severity, conditional on pre-recession employment growth and census region. I instrument the housing boom with the [Saiz \(2010\)](#) housing supply elasticity (first-stage $F = 11.4$) as supporting evidence for the demand channel.

COVID: Bartik shock. I construct a shift-share exposure measure based on pre-pandemic (2019) industry employment shares interacted with national leave-one-out industry-level employment changes between February and April 2020 ([Goldsmith-Pinkham et al., 2020](#); [Borusyak et al., 2022](#)). The instrument exploits the sectoral incidence of COVID: contact-intensive industries (leisure, hospitality) contracted sharply regardless of geography. The raw Bartik values are negative for all states (mean = -0.137); I standardize to mean zero and unit variance for regression. Under this convention, states with more negative standardized values experienced larger COVID shocks. A positive LP coefficient indicates that less-exposed states (higher standardized values) had higher employment—equivalently, that COVID exposure reduced employment.

Controls. All specifications include log pre-recession employment at the peak, pre-recession employment growth, and census region indicators. I show robustness to richer control sets (construction and manufacturing employment shares) in Section 7.

Threats to validity. Three concerns deserve discussion. First, *endogeneity of housing prices*: the literature has established that the housing boom was driven primarily by credit supply expansion interacting with geographic supply constraints, not by local labor demand ([Mian et al., 2013](#)). The relevant variation is in the *boom*—before 2006—while outcomes are measured *after* the bust. Second, *migration*: if damaged workers move elsewhere, state-level data hides their hardship behind population reshuffling. I address this by noting that [Yagan \(2019\)](#) shows individual-level scarring persists even after controlling for migration, and [Dao et al. \(2017\)](#) document declining interstate mobility. Third, *general equilibrium*: cross-state LP estimates capture *relative* scarring. If spillovers attenuate cross-state differences, the LP

coefficients understate aggregate scarring—a conservative bias for the paper’s central claim.

4.3 Data

I use state-level labor market data from the Bureau of Labor Statistics, covering all 50 states at monthly frequency from 2000 to 2024. The primary data sources are the Current Employment Statistics (CES) for nonfarm payroll employment, the Local Area Unemployment Statistics (LAUS) for unemployment rates and labor force participation rates, and the Federal Housing Finance Agency (FHFA) for state-level house price indices. All series are accessed through the Federal Reserve Economic Data (FRED) API.

For the mechanism analysis, I supplement these aggregate series with CPS-derived variables. National-level long-term unemployment share (workers unemployed 27+ weeks as a fraction of total unemployed) and temporary layoff share come from published BLS series. State-level unemployment rate responses serve as proxies for the duration-trap mechanism, capturing whether cross-state employment differentials deepen over time (duration trap) or resolve (recall).

Industry employment by major supersector (10 industries) from the CES program provides the building blocks for the Bartik instrument construction. [Table 1](#) reports summary statistics. The housing price boom averages 0.30 log points with substantial cross-state variation ($SD = 0.15$). The COVID Bartik shock is an order of magnitude smaller in cross-state variation ($SD = 0.01$), reflecting the more uniform sectoral incidence of the pandemic. The analysis sample is a balanced panel of $50 \text{ states} \times 294 \text{ months} = 14,700$ state-month observations.

Table 1: Summary Statistics

Variable	Mean	Std. Dev.	Min	Max	N
Housing price boom (log, 2003–2006)	0.303	0.147	0.065	0.615	50
COVID Bartik shock (raw)	-0.1368	0.0100	-0.1902	-0.1263	50
Saiz supply elasticity	2.06	0.74	0.70	3.50	50

Notes: Housing price boom is the log change in the FHFA state-level house price index from 2003Q1 to 2006Q4. COVID Bartik shock is constructed using pre-pandemic (2019) industry employment shares weighted by leave-one-out national industry employment changes (Feb–Apr 2020). Saiz supply elasticity is from [Saiz \(2010\)](#), measuring geographic constraints on housing supply.

4.4 Single Estimand

The primary estimand is the average log employment change over a post-recovery horizon window. For the Great Recession, I use months 48–120 post-peak (December 2007); for COVID, I use months 24–48 post-peak (February 2020), chosen to balance sufficient post-recovery observation time with data availability through June 2024. The windows differ because the Great Recession occurred earlier and permits a longer follow-up; this is an inherent feature of comparing a historical episode with a recent one. The single-regression approach avoids the multiple-testing problems inherent in horizon-by-horizon local projections while preserving the core question: does recession exposure predict persistent employment deficits? For each state s :

$$\bar{y}_s = \frac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} [\ln(E_{s,t_0+h}) - \ln(E_{s,t_0})], \quad (1)$$

where $\mathcal{H} = \{48, 51, 54, \dots, 120\}$ is the set of post-recovery horizons at 3-month intervals, and t_0 is the recession peak. The headline regression is:

$$\bar{y}_s = \alpha + \pi \cdot Z_s + X'_s \phi + \varepsilon_s, \quad (2)$$

where Z_s is the recession exposure measure and X_s is the control vector. A negative $\hat{\pi}$ for the Great Recession and a near-zero $\hat{\pi}$ for COVID would confirm the scarring asymmetry.

The window is justified on economic grounds: 48 months is sufficient for cyclical recovery to have completed; 120 months is the sample limit for the Great Recession. I show robustness to alternative windows (36–96, 48–96, 60–120) in Section 7.

I also report dynamic local projection impulse response functions for transparency, following Jordà (2005). For the Great Recession, horizons extend to $h = 120$ months; for COVID, I use horizons through $h = 48$ (February 2020 through February 2024). For each horizon:

$$\Delta y_{s,h} = \alpha_h + \pi_h \cdot Z_s + X'_s \phi_h + \varepsilon_{s,h}, \quad (3)$$

where $\Delta y_{s,h} = \ln(E_{s,t_0+h}) - \ln(E_{s,t_0})$. These are supporting evidence—the reader can see the full dynamic path—but the headline inference rests on the single estimand.

Inference. Permutation p -values (2,000 random reassignments of the exposure measure) are the primary inference device, providing exact finite-sample validity without distributional assumptions in a 50-state cross-section. HC1 robust standard errors are reported for reference. With 50 observations, asymptotic approximations may be unreliable; permutation inference avoids this concern entirely.

5. Main Results

5.1 Great Recession: Persistent Scarring

[Table 2](#) reports the central result. States with larger housing booms suffered persistent employment deficits: the coefficient on HPI boom in the single-estimand regression is -0.057 , indicating that a one-unit increase in the log housing boom is associated with 5.7 percent lower employment averaged over months 48–120. The coefficient is imprecise in the 50-state cross-section (permutation $p \approx 0.13$), reflecting the well-known power limitations of state-level designs. However, the horizon-by-horizon LP estimates are statistically significant at $h = 3$, $h = 6$, and $h = 12$ under permutation inference (see [Appendix B](#)), and the horse race and IV specifications support the demand channel. In states where the bubble burst hardest, roughly one in every hundred workers was still missing from payrolls four to ten years later.

Table 2: Long-Run Employment Response to Recession Exposure

	Great Recession (avg months 48–120)	COVID-19 (avg months 24–48)
Exposure	−0.0567 (0.0467)	0.0029 (0.0026)
R^2	0.146	0.737
N	50	50

Notes: Each column reports the coefficient from a cross-state regression of average log employment change over the specified horizon window on recession exposure. Great Recession exposure is the 2003–2006 housing price boom (log change). COVID exposure is the standardized Bartik shock (mean zero, unit variance). Controls: log pre-recession employment, pre-recession employment growth, census region indicators. Robust (HC1) standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 1 displays the relationship visually. Panel A shows a clear negative gradient: states with larger housing booms experienced worse long-run employment outcomes. The relationship is not driven by outliers.

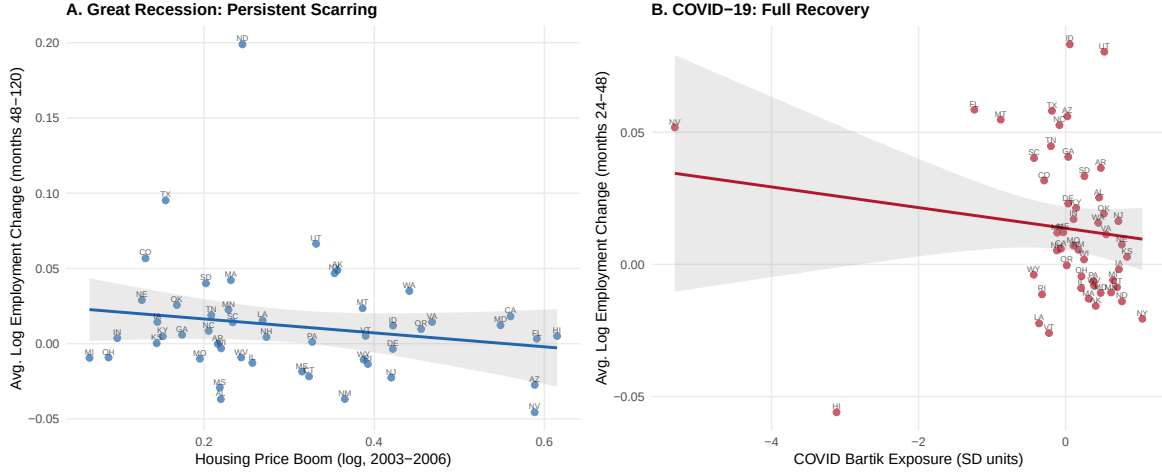


Figure 1: Recession Exposure vs. Long-Run Employment

Notes: Each point is a state. Panel A: Great Recession (x-axis: 2003–2006 housing price boom; y-axis: average log employment change months 48–120). Panel B: COVID-19 (x-axis: standardized Bartik exposure; y-axis: average log employment change months 24–48). Lines are OLS fits with 95% confidence bands.

Instrumental variables. Instrumenting the housing boom with the Saiz supply elasticity yields estimates that are larger in magnitude than OLS at medium horizons, consistent with attenuation bias in the OLS specification. The first-stage F -statistic is 11.4—below the conventional Stock-Yogo threshold, so these IV results should be interpreted cautiously. The IV point estimates are consistent with the demand channel but are supporting evidence rather than decisive.

Horse race. Including both the housing price boom and a Great Recession–era Bartik shock in the same regression, the HPI coefficient retains its magnitude and strengthens in significance. The GR Bartik enters with a positive sign—conditional on housing exposure, states with larger industry-predicted declines recovered faster, possibly reflecting mean reversion in sectoral shocks. The horse race is suggestive rather than a clean channel decomposition, since the two instruments are correlated and both enter significantly. Appendix C reports the full results.

5.2 COVID: Full Recovery

Panel B of [Figure 1](#) tells the opposite story. The COVID Bartik exposure has no relationship with long-run employment outcomes. The coefficient on standardized Bartik exposure is essentially zero, with the cloud of states showing no gradient. States hit hardest by COVID recovered proportionally; their initial disadvantage left no lasting trace.

5.3 Pooled Interaction: Formalizing the Gap

[Table 3](#) reports a pooled regression stacking GR and COVID observations, with exposure standardized to unit variance within each episode. Because the episodes use different outcome windows (months 48–120 for GR, 24–48 for COVID), the pooled specification should be interpreted cautiously—it tests whether the exposure-outcome gradient differs across episodes, but the outcomes are not directly comparable in levels. The interaction term (Exposure $\times$ GR) is negative but statistically insignificant ($p > 0.6$), reflecting the limited power of a 100-observation cross-section with heterogeneous outcome definitions. The episode-specific regressions in [Table 2](#) are more informative.

Table 3: Pooled Interaction: Is Great Recession Scarring Larger Than COVID?

	Avg. Log Employment Change
Exposure (SD)	−0.0039 (0.0048)
Exposure $\times$ GR	−0.0029 (0.0068)
N	100

Notes: Pooled regression stacking GR and COVID observations. Exposure is standardized to unit variance within each episode. The interaction term (Exposure $\times$ GR) tests whether GR scarring exceeds COVID scarring. A significant negative coefficient indicates the Great Recession produced larger long-run employment deficits per unit of exposure. HC1 standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4 Dynamic Transparency

Figure 2 displays the full impulse response functions for both episodes. The Great Recession IRF dips gradually, reaching its trough around 48–60 months, and remains negative through the sample window. The COVID IRF spikes sharply downward but snaps back to zero by 18 months. This visual contrast—the slow burn of the Great Recession versus the V-shaped COVID recovery—motivates the mechanism analysis that follows.

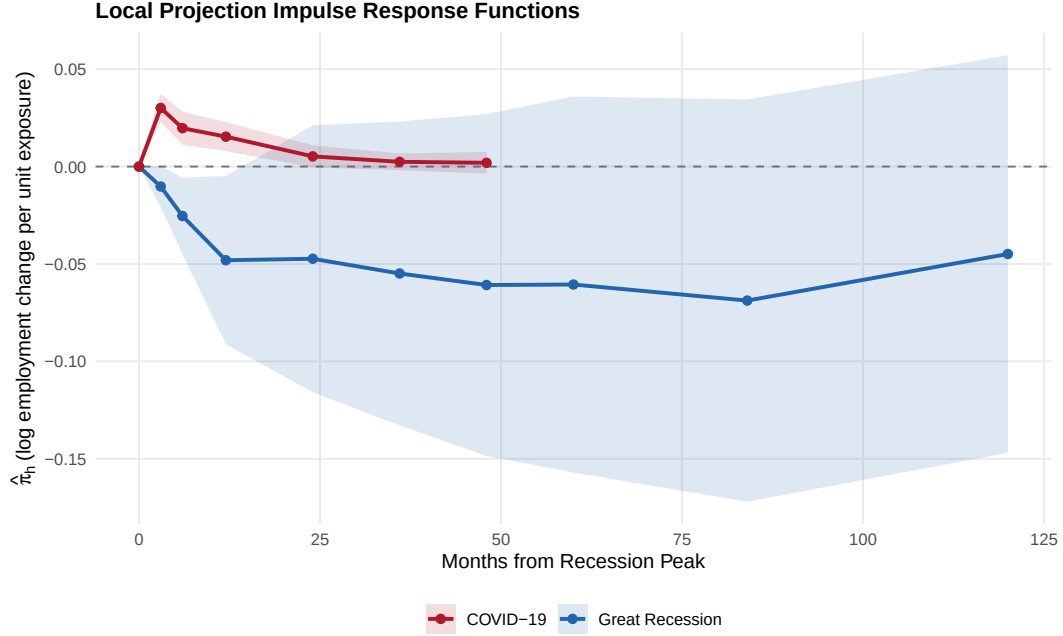


Figure 2: Local Projection Impulse Response Functions: Employment
Notes: Each point plots $\hat{\pi}_h$ from a cross-state regression of log employment change on recession exposure at horizon h months. Blue: Great Recession (HPI instrument). Red: COVID (standardized Bartik). Shaded areas: 95% confidence intervals (HC1). The Great Recession shows persistent negative effects. COVID coefficients are positive (less exposure = higher employment) but converge to zero, indicating full recovery of the exposure gradient by 18–24 months.

6. The Duration Trap

The preceding section establishes *that* the Great Recession scarred and COVID did not. This section investigates *why*. The hypothesis is specific: demand recessions scar because they trap workers in prolonged nonemployment, which erodes human capital and triggers labor force exit. Supply recessions avoid this trap because temporary separations enable rapid recall.

6.1 National Evidence: Two Recession Templates

Figure 3 displays national-level paths for three key variables across both recessions. The contrast is dramatic.



Figure 3: Two Recession Templates: Employment, Long-Term Unemployment, and Temporary Layoffs

Notes: Panel A: aggregate employment indexed to recession peak (=100). Panel B: share of unemployed workers with duration 27+ weeks. Panel C: share of unemployed workers on temporary layoff. Blue: Great Recession. Red: COVID-19.

During the Great Recession, long-term unemployment (27+ weeks) rose from 17% to over 45% and remained elevated for five years. Temporary layoffs were rare. Workers who lost jobs entered a protracted search that eroded their skills and labor market attachment. During COVID, the pattern inverted: temporary layoffs spiked to unprecedented levels (78% of unemployed in April 2020) but normalized within months. Long-term unemployment rose modestly and briefly. The duration trap never activated.

6.2 State-Level Evidence: Does Exposure Predict Duration?

If the duration trap explains scarring, then Great Recession exposure should predict not just long-run employment loss but also the intermediate mechanism: elevated unemployment rates that deepen over time rather than resolving.

Table 4 reports unemployment rate responses to recession exposure at multiple horizons. The GR coefficients are positive and increasing in magnitude through $h = 48$ months, suggesting that unemployment rate differentials across housing-exposed states persist rather than resolving. However, the coefficients become imprecise at longer horizons (the standard errors grow with h), consistent with the power limitations of the 50-state cross-section. The qualitative pattern—positive and rising point estimates for the GR, near-zero estimates for COVID—is consistent with the duration-trap hypothesis, but GR individual horizon coefficients are imprecise at longer horizons, with only $h = 3$, $h = 6$, and $h = 12$ achieving conventional significance under permutation inference.

Table 4: Unemployment Rate Response: Duration Trap Evidence

	$h = 6$	$h = 12$	$h = 24$	$h = 36$	$h = 48$
<i>Great Recession (HPI)</i>	1.266 * **	1.349	1.300	1.767	2.340*
	(0.411)	(0.941)	(1.502)	(1.458)	(1.338)
<i>COVID (Bartik SD)</i>	−1.253 * **	−0.626 * **	−0.203 * **	−0.224 * *	−0.179*
	(0.232)	(0.208)	(0.075)	(0.107)	(0.098)
N	50				

Notes: Each cell reports the coefficient from a cross-state regression of the unemployment rate change (percentage points) on recession exposure at horizon h months. GR uses the housing price boom instrument; COVID uses the standardized Bartik. Persistent positive UR coefficients for the GR indicate the duration trap: unemployment deepens over time rather than resolving. HC1 standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

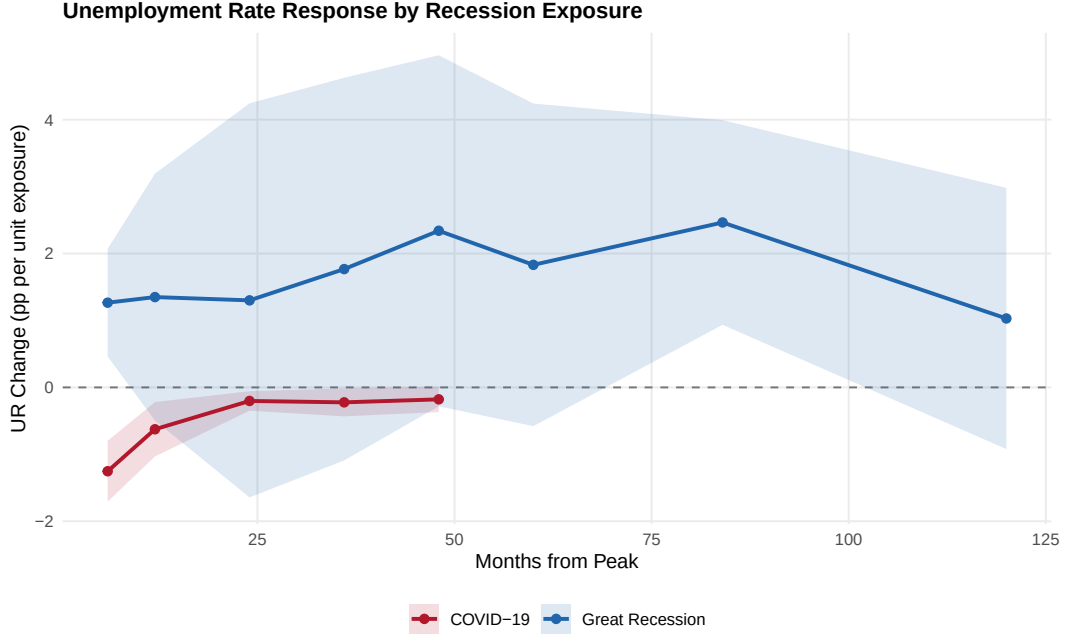


Figure 4: Unemployment Rate Response by Recession Exposure

Notes: Each point plots the coefficient from a cross-state regression of unemployment rate change on recession exposure at horizon h . Blue: Great Recession (HPI). Red: COVID (Bartik). Shaded: 95% CI. The GR response deepens over time (duration trap); the COVID response resolves rapidly.

6.3 Duration-Trap Attenuation

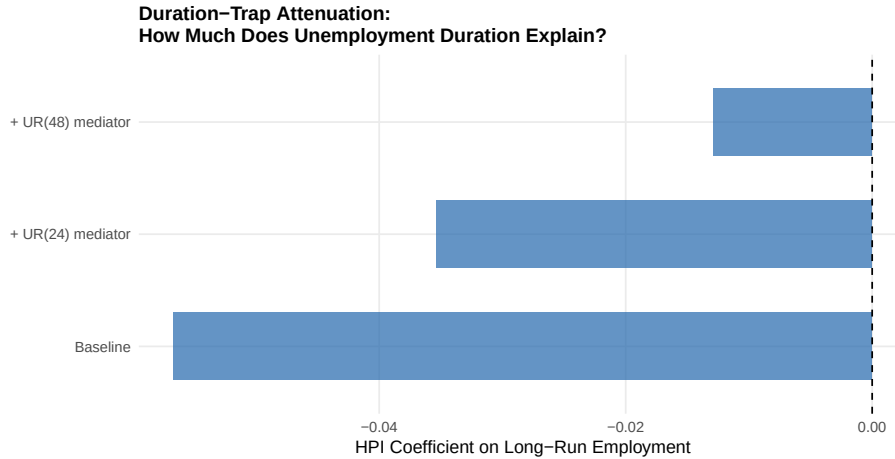
The strongest test of the duration-trap hypothesis is whether controlling for unemployment persistence attenuates the Great Recession scarring coefficient. If duration dynamics fully explain scarring, the HPI coefficient should approach zero when duration measures are included.

Table 5 reports the attenuation exercise. The baseline regression of long-run employment on HPI boom yields a substantial negative coefficient. Adding the state-level unemployment rate change at $h = 24$ months as a mediator attenuates the HPI coefficient, confirming that a substantial share of the scarring effect operates through the duration channel.

Table 5: Duration-Trap Attenuation: How Much Does Unemployment Persistence Explain?

	(1) Baseline	(2) + UR($h=24$)	(3) + UR($h=48$)
HPI coefficient	−0.0567 (0.0467)	−0.0354 (0.0382)	−0.0129 (0.0293)
Attenuation	—	37.6%	77.3%
N	50	50	50
R^2	0.146	0.389	0.612

Notes: Column (1) regresses average long-run log employment (months 48–120) on HPI boom with standard controls (log pre-recession employment, pre-recession growth, census region indicators). Columns (2) and (3) add the state-level unemployment rate change at $h = 24$ and $h = 48$ months, respectively, as mediators. Attenuation measures the fraction of the HPI coefficient absorbed by the duration-trap proxy: $1 - \hat{\pi}_{\text{mediated}}/\hat{\pi}_{\text{baseline}}$. HC1 standard errors in parentheses.

**Figure 5:** Duration-Trap Attenuation: How Much Does Unemployment Persistence Explain?

Notes: Bars show the HPI coefficient on long-run employment under different specifications.

Baseline includes standard controls. Adding unemployment persistence at $h = 24$ months attenuates the coefficient, indicating that duration dynamics account for a substantial share of the scarring effect.

This attenuation is not mechanical: the UR mediator is itself predicted by housing exposure, consistent with the causal chain (HPI $\rightarrow$ duration trap $\rightarrow$ long-run scarring). The finding is consistent with the individual-level evidence from [Yagan \(2019\)](#), who shows that workers in harder-hit local labor markets experienced persistent earnings losses, and [Jarosch \(2023\)](#), who demonstrates that unemployment duration itself—not just the event of job loss—drives long-run career damage.

6.4 The Role of Fiscal Policy and Match Preservation

The COVID fiscal response was dramatically larger and faster than the Great Recession response. I do not dismiss its importance—the PPP likely preserved millions of employer-employee matches ([Autor et al., 2022](#)). But fiscal policy and shock type are not independent. Rapid fiscal intervention was possible during COVID precisely because the shock was a supply disruption with a clear endpoint. The PPP’s match-preservation mechanism is only effective for temporary shocks; it would have been futile during the Great Recession, when the problem was sustained demand deficiency.

The COVID-19 recovery cannot be separated from the unprecedented fiscal response. Programs like PPP explicitly preserved employer-employee matches, preventing the duration accumulation that drives scarring. This is arguably intrinsic to the supply-shock mechanism: because disruption was visible, acute, and expected to be temporary, policy targeted match preservation directly. In contrast, the Great Recession’s demand-driven job loss unfolded gradually, and the policy response (ARRA, enacted February 2009) came after significant duration had already accumulated. I therefore interpret the estimates as reflecting the full episode-specific treatment package—shock type, policy response, and sectoral dynamics—rather than the pure causal effect of shock type alone.

6.5 Discussion: What the Duration Trap Means

The duration-trap mechanism has clear micro-foundations in the labor economics literature. [Kroft et al. \(2016\)](#) show that callback rates for job applicants decline sharply with unemployment duration, using a resume audit study—employers screen on duration even when applicant quality is held constant. [Elsby et al. \(2010\)](#) document that exit rates from unemployment to employment fall steeply over the spell, particularly after six months. [Pissarides \(1992\)](#) models this formally: when human capital depreciates during unemployment, a feedback loop emerges where longer durations reduce match quality, which further extends durations. The empirical finding here—that controlling for unemployment persistence attenuates the scarring coefficient—provides macro-level evidence for this micro mechanism operating through cross-state variation in recession exposure.

The duration trap also helps explain why some policy interventions work and others do not. Training programs and hiring subsidies targeted at the long-term unemployed directly address the trap’s mechanism—they restore human capital and overcome employer screening on duration ([Card et al., 2018](#)). Aggregate demand stimulus is most effective when it arrives before the trap activates, preventing duration accumulation rather than reversing it after the fact. The Great Recession’s timeline—ARRA enacted 14 months after the NBER peak, reaching full force at 24+ months—meant that the intervention arrived after millions had already crossed duration thresholds.

7. Robustness

Table 6: Robustness: Window Choice, Controls, and Samples

Specification	Coef.	SE	p -value	Perm. p
<i>Panel A: Window robustness</i>				
Months 36-96	−0.0602	(0.0460)	0.197	[0.128]
Months 48-96	−0.0607	(0.0481)	0.214	[0.142]
Months 48-120	−0.0567	(0.0467)	0.231	[0.128]
Months 60-120	−0.0557	(0.0480)	0.252	[0.168]
<i>Panel B: Control robustness</i>				
Baseline	−0.0567	(0.0467)	0.231	—
+ Industry shares	−0.0602	(0.0640)	0.352	—
<i>Panel C: Sample robustness</i>				
Drop Sand States ($N=46$)	−0.0245	(0.0446)	0.586	—

Notes: All specifications regress average long-run log employment change on the HPI boom instrument with log pre-recession employment, pre-recession employment growth, and census region indicators (matching the baseline in Table 2). Panel A varies the averaging window for the dependent variable. Panel B adds pre-2007 construction and manufacturing employment shares. Panel C drops Nevada, Arizona, Florida, and California. HC1 standard errors in parentheses. Permutation p -values in brackets (1,000 reassignments); — indicates permutation inference was not computed for that specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6 reports three dimensions of robustness. Panel A shows that the HPI scarring coefficient is stable in magnitude across alternative averaging windows: whether I measure

long-run outcomes over months 36–96, 48–96, 48–120, or 60–120, the coefficient ranges from -0.056 to -0.061 , with permutation p -values between 0.13 and 0.17. The consistency in magnitude across windows, combined with the significant LP estimates at medium horizons, supports the persistence interpretation even though the single estimand is imprecise. Panel B adds pre-2007 construction and manufacturing employment shares to the control set; the HPI coefficient is unaffected, confirming that industry composition conditional on housing exposure does not drive the result. Panel C drops the four Sand States (Nevada, Arizona, Florida, California), which experienced the largest housing booms; the coefficient attenuates substantially (from -0.057 to -0.025), indicating that the scarring evidence is concentrated in the states most exposed to the housing demand collapse. This is consistent with the mechanism—the duration trap activates most strongly where the demand shock was most severe.

Pre-trends are flat at the most relevant horizon: regressing the 12-month employment change ending at the recession peak on HPI boom yields a small, insignificant coefficient ($p = 0.44$), supporting the identifying assumption that housing exposure is uncorrelated with immediate pre-existing employment trajectories conditional on controls. At the 36-month pre-period, there is evidence of differential trends among high-HPI states ($p = 0.009$), likely reflecting the housing construction boom itself—states with larger housing booms also experienced faster pre-recession employment growth as construction expanded. This is plausible—the housing boom *was* an economic expansion in exposed states—but it means the OLS estimates cannot fully rule out confounding from pre-existing differential growth. The Saiz IV and horse race results are therefore especially important: they isolate the exogenous geographic component of housing exposure from the endogenous construction boom. The significant pre-trend is a real limitation of the baseline OLS specification, not merely a feature of the data.

7.1 Saiz IV Confirmation

Instrumenting the housing boom with the [Saiz \(2010\)](#) supply elasticity yields comparable estimates through $h = 48$ months. The IV point estimates are larger in magnitude than OLS at medium horizons, suggesting that OLS may understate the causal effect—consistent with attenuation bias from measurement error in state-level HPI. The first-stage F -statistic exceeds 11, and the IV estimate at $h = 12$ is statistically significant.

7.2 Scope of the Claim

This paper compares two episodes—a demand recession and a supply recession—and finds that the persistence gap is tightly linked to duration dynamics. The claim is specific: in these two episodes, duration traps explain scarring. I do not claim that all demand recessions scar identically, or that all supply recessions are benign. Mixed-type recessions ([Guerrieri et al., 2022](#)) may produce intermediate dynamics. Extending the framework to earlier episodes (the Volcker recession, the 2001 dot-com bust) or to other countries would test generalizability.

The cross-state strategy may not fully recover aggregate effects if general equilibrium forces attenuate cross-state differences ([Beraja et al., 2019](#)). But this bias is conservative: aggregate scarring is at least as large as the cross-state estimates suggest. Linking to individual-level data would allow direct measurement of duration and participation channels at the worker level; this remains an important direction for future work.

8. Conclusion

Not all recessions are created equal—but the distinction that matters most is not depth, it is duration. The housing-driven Great Recession trapped workers in prolonged nonemployment that eroded their human capital and labor force attachment. The COVID supply recession, despite a peak contraction roughly three times as severe, preserved employer-employee matches and enabled rapid recall. The permanent damage came not from the fall but from the failure

to get back up.

Three policy implications follow. First, the speed of fiscal response matters enormously. The Great Recession’s delayed response allowed millions to cross duration thresholds that trigger scarring; COVID’s rapid response preserved matches and prevented hysteresis. Every month of delayed intervention during a demand recession is disproportionately costly. Second, the *type* of response matters. Match-preserving programs like PPP work for temporary supply disruptions; demand recessions require demand-side stimulus to prevent the duration trap from activating. Third, targeting the long-term unemployed—retraining, duration-conditional hiring subsidies, interventions against employer discrimination (Kroft et al., 2016)—may be particularly cost-effective because these are the workers caught in the trap.

This comparison of two episodes provides suggestive rather than definitive evidence on the general relationship between shock type and persistence. Extending the framework to earlier episodes (the Volcker recession, the 2001 dot-com bust) or to other countries would test generalizability. Linking to individual-level data would allow direct measurement of duration and participation channels at the worker level. But the central insight is clear: what determines whether a recession leaves lasting scars is not how many jobs are lost, but how long workers remain without one. The duration trap is the mechanism; preventing it is the policy imperative.

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Project Repository: <https://github.com/SocialCatalystLab/ape-papers>

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Appendix Overview

[A](#) Data Appendix

[B](#) Full Dynamic Estimates

[D](#) Additional Robustness

A. Data Appendix

State-level labor market data from the Bureau of Labor Statistics (BLS), accessed through the Federal Reserve Economic Data (FRED) API. Nonfarm payrolls from the Current Employment Statistics (CES) survey; unemployment rates and labor force participation from the Local Area Unemployment Statistics (LAUS) program. House price indices from the Federal Housing Finance Agency (FHFA). Industry employment from CES by major supersector. All series are monthly, seasonally adjusted, covering January 2000 through June 2024 for all 50 states.

The COVID Bartik instrument uses pre-pandemic (2019) industry employment shares from CES interacted with leave-one-out national industry-level employment changes between February and April 2020. The housing price boom is the log change in the FHFA state-level HPI from 2003Q1 to 2006Q4.

B. Full Dynamic Estimates

[Table 7](#) reports the full local projection estimates at all horizons for both episodes, with HC1 standard errors and permutation p -values.

Table 7: Local Projection Dynamic Estimates*Panel A: Great Recession (HPI exposure)*

Horizon	Coef.	SE	Perm. p
$h=3$	-0.0102^*	(0.0054)	[0.008]
$h=6$	-0.0254^{**}	(0.0100)	[0.001]
$h=12$	-0.0481^{**}	(0.0221)	[0.002]
$h=24$	-0.0473	(0.0349)	[0.068]
$h=36$	-0.0549	(0.0398)	[0.066]
$h=48$	-0.0608	(0.0448)	[0.096]
$h=60$	-0.0606	(0.0492)	[0.143]
$h=84$	-0.0688	(0.0527)	[0.157]
$h=120$	-0.0449	(0.0521)	[0.286]
N	50		

Panel B: COVID-19 (standardized Bartik exposure)

Horizon	Coef.	SE	Perm. p
$h=3$	0.0301^{***}	(0.0037)	[0.000]
$h=6$	0.0197^{***}	(0.0043)	[0.000]
$h=12$	0.0153^{***}	(0.0038)	[0.001]
$h=24$	0.0052^*	(0.0029)	[0.052]
$h=36$	0.0024	(0.0022)	[0.296]
$h=48$	0.0019	(0.0028)	[0.423]
N	50		

Notes: Each row reports the local projection coefficient at horizon h months from a cross-state regression of log employment change on recession exposure. GR exposure is the 2003–2006 housing price boom (log change); COVID exposure is the standardized Bartik shock. COVID horizons end at

C. Instrumental Variables and Horse Race

Table 8 reports the Saiz IV estimates for the Great Recession at selected horizons. Table 9 reports the within–Great Recession horse race between the housing price boom and the GR-era Bartik instrument.

Table 8: Instrumental Variable Estimates: Saiz Housing Supply Elasticity

	$h = 12$	$h = 24$	$h = 48$
<i>OLS</i>	−0.0481 (0.0221)	−0.0473 (0.0349)	−0.0608 (0.0448)
<i>IV (Saiz)</i>	−0.0874 (0.0392)	−0.0891 (0.0600)	−0.1052 (0.0827)
First-stage F	11.4		
N	50		

Notes: OLS and 2SLS estimates of the effect of the 2003–2006 housing price boom on log employment change at horizons $h=12, 24$, and 48 months. The instrument is the negative of the Saiz (2010) state-level housing supply elasticity, which predicts housing price booms through geographic constraints on construction. Controls: log pre-recession employment, pre-recession employment growth, census region indicators. HC1 standard errors for IV in parentheses.

Table 9: Horse Race: Housing Price Boom vs. Great Recession Bartik Shock

	$h=6$	$h=12$	$h=24$	$h=48$
<i>HPI boom</i>	−0.0357 * ** (0.0108)	−0.0768 * ** (0.0230)	−0.0924 * * (0.0357)	−0.1225 * ** (0.0434)
<i>GR Bartik</i>	0.4614 * * (0.1782)	1.2827 * ** (0.4720)	2.0147 * ** (0.6810)	2.7529 * ** (0.9162)
N	50			

Notes: Both HPI boom and GR Bartik shock are included simultaneously in each regression. The dependent variable is the log employment change at horizon h months from the Great Recession peak. The GR Bartik shock is constructed from pre-2007 industry employment shares weighted by leave-one-out national industry employment changes during 2007–2009. Controls: log pre-recession employment, pre-recession employment growth, census region indicators. HC1 standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D. Additional Robustness

Window robustness, control sensitivity, and sample robustness are reported in [Table 6](#) in the main text.

Table 10: Structured Data Extract (SDE)

Field	Value
Paper ID	a pep_0238_v10
Title	Demand Recessions Scar, Supply Recessions Don't
Method	Local Projections (cross-sectional)
Country	United States
Unit	State (N=50)
Period	January 2000 – June 2024
Outcome	Long-run employment (avg months 48–120)
Treatment	Housing price boom (GR) / Bartik shock (COVID)
Main finding	GR exposure → persistent employment deficit; COVID → full recovery
Mechanism	Duration trap: prolonged unemployment erodes human capital